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How Much AI Is in This Track?

Quantifying the proportion of AI-generated stems in hybrid music mixtures

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Audio, code & video
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Detectors assume all-human or all-AI

Detectors [1, 2] are near-perfect on pure tracks. But producers now mix AI stems with human recordings [3–5], and no binary label fits that track.

Drums · human | Bass · AI
Guitar · human | Vocals · AI

0 or 1?

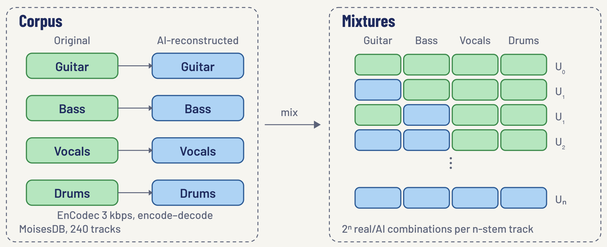
From "is it AI?" to "how much AI?"

$$\alpha = \frac{\sum_{s \in \mathcal{A}} \mathrm{RMS}(s)^2}{\sum_{\text{all } s} \mathrm{RMS}(s)^2}$$

The share of the mix's energy that comes from AI stems ($\mathcal{A}$). **α = 0** is fully human, **α = 1** fully AI.

ARIA: hybrid mixtures with known α

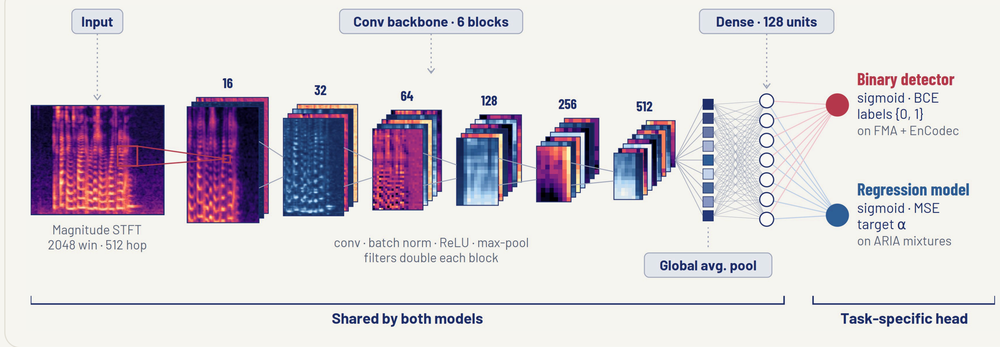
Encode–decode real MoisesDB [6] stems with EnCodec [7] at 3 kbps: same music, only the codec artifacts differ.



240 tracks **21,212** mixtures **1,792** held-out test

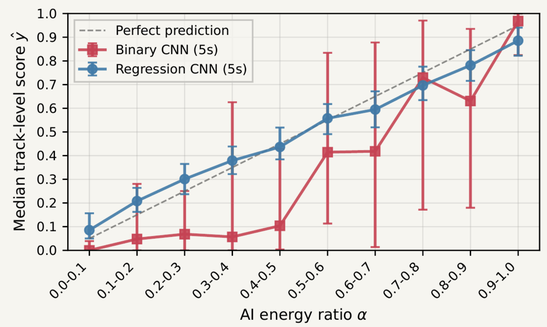
Same backbone, different head

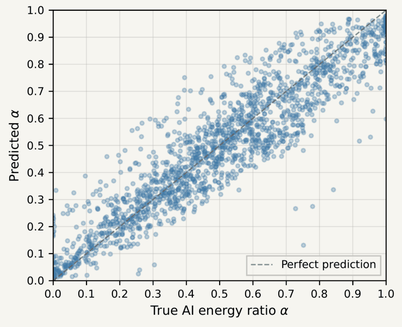
Backbone of the reproduced detector [1]; only the output layer and loss change. **Binary**: labels {0, 1} on FMA [8] + EnCodec. **Regression**: target α on ARIA. Window scores are averaged per track.



Training on α recovers the proportion

The binary detector stays near 0 until about half of the energy is AI, with a wide spread. The same backbone trained on α follows the diagonal.



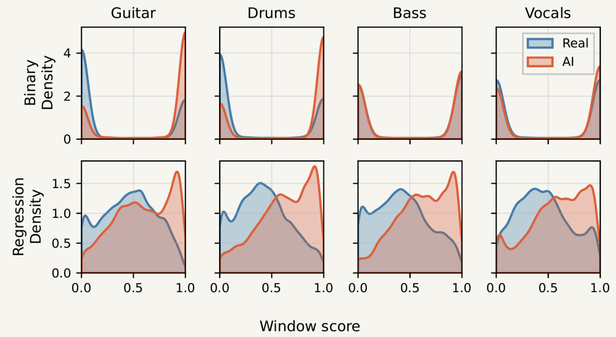


R^2 −0.85 binary score used as α: worse than predicting the mean

MAE 0.076 · R^2 0.85

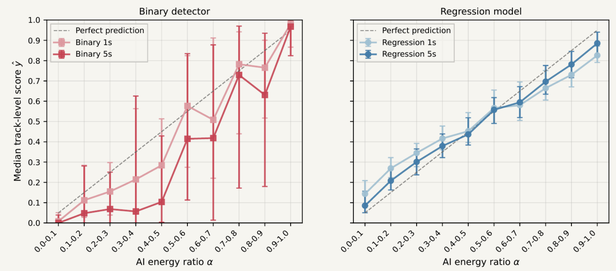
Held-out test split, 5 s windows. Left: median score per α bin, bars show the IQR. Right: regression, one point per mixture.

Sensitivity depends on the stem



Drums, guitar: binary separates real and AI cleanly.
Vocals, bass: they overlap. An AI bassline can go unnoticed.
Regression shifts AI scores right for all four.

Why 5-second windows?



Longer windows help **regression**, hurt **binary**.

Takeaways

1. Producers mix AI stems with human recordings.
2. Binary detection can't say how much AI is in a mix.
3. Estimating α directly is a promising first step.

[1] Afchar et al., "AI-generated music detection and its challenges," ICASSP 2025
[2] Rahman et al., "SONICS: Synthetic or not – identifying counterfeit songs," ICLR 2025
[3] Nistal et al., "Diff-A-Riff: Musical accompaniment co-creation via latent diffusion models," ISMIR 2024
[4] Pasini et al., "LiveBand: Live accompaniment generation in the audio domain," arXiv 2026
[5] Pons et al., "Music and artificial intelligence: Artistic trends," arXiv 2025
[6] Pereira et al., "MoisesDB: A dataset for source separation beyond 4-stems," ISMIR 2023
[7] Défossez et al., "High fidelity neural audio compression," TMLR 2023
[8] Defferrard et al., "FMA: A dataset for music analysis," ISMIR 2017

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